

STAT 432 Homework 2

Question 1: Training and test error under a fixed design

(a) Simulation + monotonicity check

Training MSE never increased within any of the 200 runs - the largest observed “increase” was -1.6×10^{-9} (floating-point noise, effectively zero), and 0 of 4,000 (run, step) pairs increased by more than 1×10^{-8} .

Why this must hold: Each model with m predictors is nested inside the model with $m + 1$ predictors - the design matrix $\mathbf{X}_m$'s column space is a subspace of $\mathbf{X}_{m+1}$'s. Least squares finds the closest point (in Euclidean distance) to $\mathbf{y}$ within that column space. Since the larger subspace contains everything the smaller one does, the best approximation to $\mathbf{y}$ found in the larger space can never be farther from $\mathbf{y}$ than the best approximation found in the smaller one. So $\text{MSE}_{\text{train},m}$ is guaranteed non-increasing in m , for every fixed $\mathbf{y}$ - this has nothing to do with expectations or repeated sampling; it's a deterministic fact about projections.

(b) Theoretical expectations vs. simulation

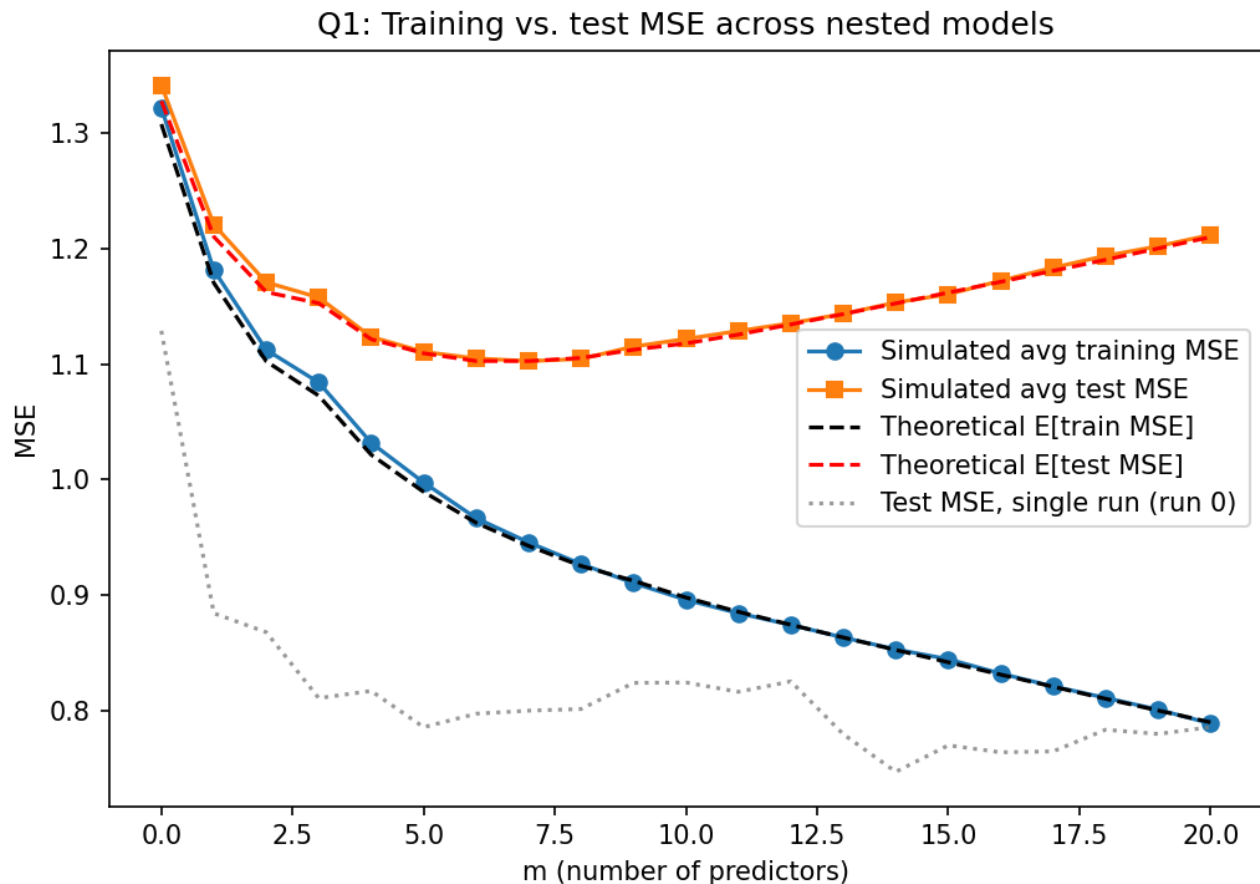


Figure 1: Q1 plot

The simulated averages track $E[\text{train}]$ and $E[\text{test}]$ closely at every m - the blue/black dashed curves overlap almost exactly, and the orange/red dashed curves do too. This is exactly what you'd expect with 200 replications averaging out the sampling noise in ϵ and ϵ^* .

The average test MSE curve dips to a minimum around $m \approx 7$ before rising again - this is the bias-variance

trade-off made visible: for small m , B_m^2/n (omitted-predictor bias) dominates and shrinks quickly as predictors are added; past some point, the true $\beta_j = 0.4\sqrt{j}$ coefficients are small enough that additional predictors buy almost no further bias reduction, so the $(m+1)/n$ estimation-variance penalty takes over and average test MSE climbs.

(c) Single test-MSE run vs. average; why training MSE alone can't select m

The single-run (gray dotted) curve is visibly noisier/jumpier than the 200-run average - it's one realization of $\text{MSE}_{\text{test},m} = \frac{1}{n}\|\epsilon^* + (\mu - \hat{\mathbf{y}}_m)\|_2^2$, so it inherits all the sampling variability of that particular ϵ^* draw and that particular $\hat{\mathbf{y}}_m$ (via ϵ). Averaging over 200 runs cancels that noise and leaves the smooth deterministic curve $B_m^2/n + 1 + (m+1)/n$.

Training MSE, visible in the plot, decreases essentially monotonically all the way to $m = 20$ - it only reflects the shrinking bias term B_m^2/n and never incorporates the $+(m+1)/n$ estimation-variance penalty that makes test MSE turn back upward. So if you picked m by minimizing training MSE alone, you'd always be pushed toward the largest available model ($m = 20$), even though the plot shows the actual expected out-of-sample performance is best around $m \approx 7$ - training error is systematically blind to the overfitting cost that variance imposes on new data.

Question 2: Prediction error at one target point

(a) Theoretical bias/variance decomposition at $\mathbf{x}_0$

$$\mu_0 = \mathbf{x}_0^\top \beta = 0(0.5) + 1(0.5) + 0(0.5) + 0(0.5) + 1(0.5) + 0(0.5) = 1.0$$

m	bias ²	$E[(\hat{\mu}_{0,m} - \mu_0)^2]$
0, 1	1.00	1.01
2, 3, 4	0.25	0.27
5, 6	0.00	0.03

Smallest m minimizing theoretical error: **$m = 5$** .

Write-up: The bias term $\left(\sum_{j=m+1}^p x_{0j}\beta_j\right)^2$ only changes value when the running sum crosses one of the two nonzero coordinates of $\mathbf{x}_0$ (positions 2 and 5). Going from $m = 0$ to $m = 1$ doesn't touch it, because predictor 1 has $x_{0,1} = 0$ - it's not part of μ_0 regardless of whether it's in the model. The bias drops from 1.00 to 0.25 exactly at $m = 2$ (predictor 2, with $x_{0,2} = 1$, has now moved from "omitted" to "included"), stays flat through $m = 3, 4$ (predictors 3 and 4 are again irrelevant to $\mathbf{x}_0$), and drops to 0 exactly at $m = 5$ (predictor 5 enters). It's flat again at $m = 6$ since predictor 6 doesn't touch $\mathbf{x}_0$ either.

The variance term $\frac{\sigma^2}{n} \left(1 + \sum_{j=1}^m x_{0j}^2\right)$ moves in lockstep for the same reason, just in the opposite direction - it only grows when an *included* predictor has $x_{0j} \neq 0$, so it also only changes at $m = 2$ and $m = 5$. Since the bias drop dominates the small variance increase at each of those steps, expected error falls at $m = 2$ and $m = 5$ and is flat everywhere else - matching the plot's staircase shape exactly.

(b) Simulation vs. theory

The 200-rep simulated averages track the theoretical curve closely at every m (blue solid vs. black dashed in the plot), with the same flat segments and the same two drop points at $m = 2$ and $m = 5$.

(c) Why only predictors 2 and 5 matter at $\mathbf{x}_0$; contrast with Q1

Because $\mathbf{x}_0 = (0, 1, 0, 0, 1, 0)^\top$, only coordinates 2 and 5 are nonzero, so $\mu_0 = x_{0,2}\beta_2 + x_{0,5}\beta_5$ - predictors 1, 3, 4, and 6 multiply a zero and drop out of μ_0 entirely, even though all six $\beta_j = 0.5$ are nonzero and matter for predicting *other* rows of the design. This is the key contrast with Question 1's test MSE: that quantity averages squared error over *all* rows of $\mathbf{X}_{\text{all}}$, so every predictor with nonzero β_j contributes to *some* row and hence to the

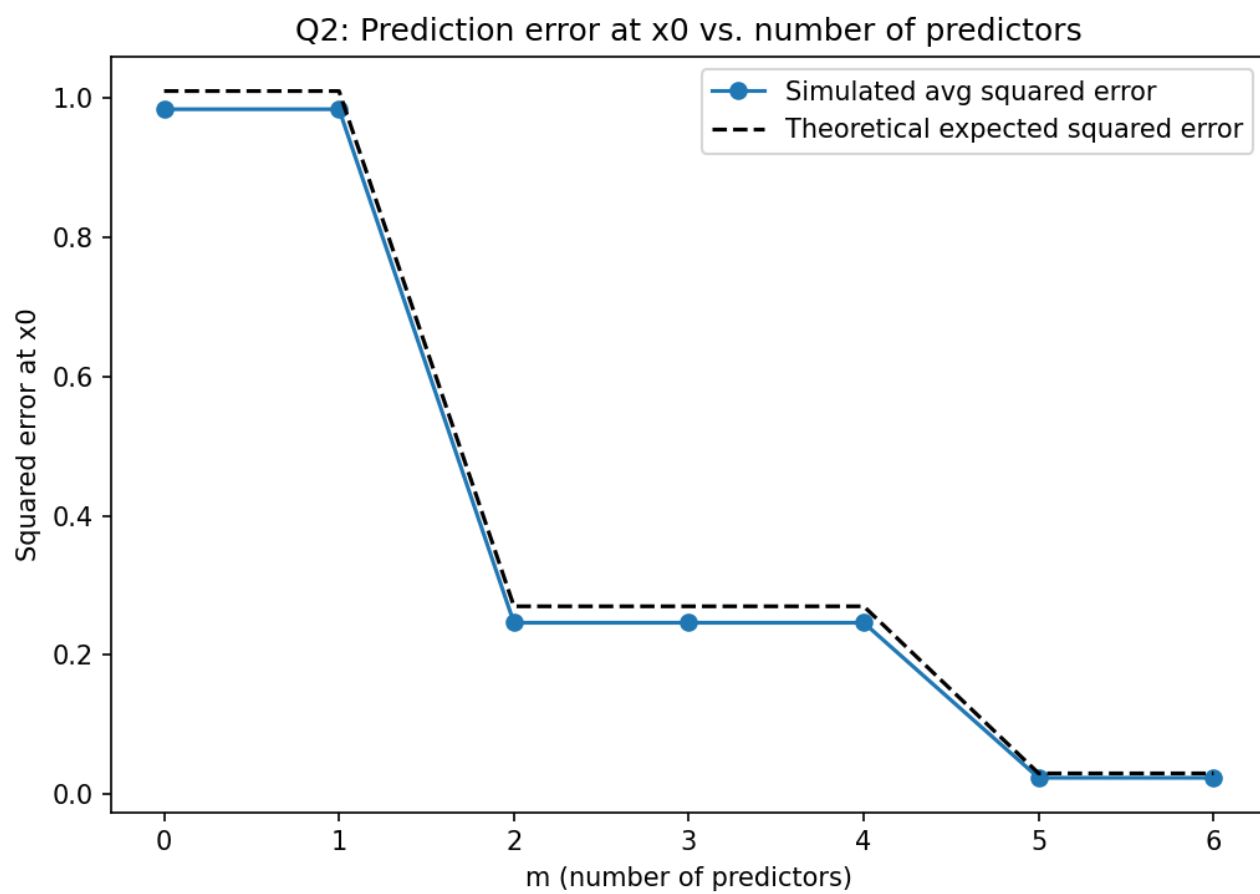


Figure 2: Q2 plot

average - which is why Q1's expected test MSE improved gradually and smoothly as m grew from 0 to 20, rather than jumping only at two specific values of m .

The practical upshot: the m that's best for predicting the mean response at one particular covariate value ($m = 5$ here, or arguably $m = 2$ if you only care about getting *most* of the benefit cheaply) can differ from the m that's best for average test performance over the whole design (Q1's $m \approx 7$) - precisely because a single target point only "sees" the subset of predictors active at that point, while average test MSE sees all of them.

Question 3: The optimism correction

(a) Derivation

Write $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$. Since $(\mathbf{I}_n - \mathbf{H})$ is idempotent and symmetric, decompose:

$$\mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu} + (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}, \quad \mathbf{y}^* - \mathbf{H}\mathbf{y} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu} + \boldsymbol{\epsilon}^* - \mathbf{H}\boldsymbol{\epsilon}.$$

Training error. $n \cdot E(\text{MSE}_{\text{train}} \mid \mathbf{X}) = \|(\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu}\|^2 + E[\boldsymbol{\epsilon}^\top (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon} \mid \mathbf{X}]$ - the cross term vanishes since $(\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu}$ is fixed given $\mathbf{X}$ and $E[\boldsymbol{\epsilon}] = 0$. Using $E(\boldsymbol{\epsilon}^\top \mathbf{A} \boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \text{tr}(\mathbf{A})$ with $\mathbf{A} = \mathbf{I}_n - \mathbf{H}$, whose trace is $n - (p + 1)$:

$$E(\text{MSE}_{\text{train}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2 \left(1 - \frac{p + 1}{n}\right).$$

Test error. $\boldsymbol{\epsilon}^*$ and $\boldsymbol{\epsilon}$ are independent, so $n \cdot E(\text{MSE}_{\text{test}} \mid \mathbf{X}) = B^2 + E[\boldsymbol{\epsilon}^{*\top} \boldsymbol{\epsilon}^*] + E[\boldsymbol{\epsilon}^\top \mathbf{H}^\top \mathbf{H} \boldsymbol{\epsilon}]$ (cross terms vanish by independence/zero mean). The first term is $\sigma^2 n$; using $\mathbf{H}^\top \mathbf{H} = \mathbf{H}^2 = \mathbf{H}$ and $\text{tr}(\mathbf{H}) = p + 1$, the second is $\sigma^2(p + 1)$. Dividing by n :

$$E(\text{MSE}_{\text{test}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2 \left(1 + \frac{p + 1}{n}\right).$$

(b) Expected optimism and numeric example

For $n = 120$, $p = 4$, $B^2/n = 0.04$, $\sigma^2 = 1$:

- $E[\text{train MSE}] = 0.99833$
- $E[\text{test MSE}] = 1.08167$
- Expected optimism $= 0.08333 = \frac{2(p + 1)}{n} = \frac{2 \cdot 5}{120}$

Write-up: $E(\text{MSE}_{\text{test}} - \text{MSE}_{\text{train}} \mid \mathbf{X}) = \sigma^2 \cdot \frac{2(p+1)}{n}$ - the bias term B^2/n cancels exactly, leaving a purely variance-driven optimism that grows with model size p and shrinks with n . This ordering ($E[\text{test}] > E[\text{train}]$) is a statement about *expectations*: for any single realized $(\mathbf{y}, \mathbf{y}^*)$ pair, $\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}^*$ are random draws, so it's entirely possible for a particular test draw to land closer to $\hat{\mathbf{y}}$ than the training draw did - the 0.08333 gap is only guaranteed on average, not realization by realization (consistent with the single noisy test-MSE curve in Question 1 jumping around from its expectation).

(c) C_p and the corrected test MSE estimate

With $\text{RSS} = 116.4$, $\hat{\sigma}^2 = 0.96$: $\widehat{\text{MSE}}_{\text{test}} = 1.05000$, $C_p = 11.25000$, and $\hat{\sigma}^2(C_p + n) = 126.00000$ matches $n \cdot \widehat{\text{MSE}}_{\text{test}} = 126.00000$ exactly.

Write-up: This confirms $\widehat{\text{MSE}}_{\text{test}} = \frac{\hat{\sigma}^2(C_p + n)}{n}$, i.e. C_p is an affine, strictly increasing transformation of $\widehat{\text{MSE}}_{\text{test}}$ whenever n and $\hat{\sigma}^2$ are the same fixed constants across every candidate model being compared. An affine increasing transform preserves ordering, so whichever candidate model minimizes one criterion also minimizes the other - they're interchangeable for model *selection* purposes even though their numeric scales differ.

Question 4: Comparing Mallows' C_p , AIC, and BIC

(a) Fitted models

- Full model ($p=10$): $RSS = 1,089,451.7212$, residual $df = 359$, $\hat{\sigma}^2 = 3034.6845$
- Model A ($p=7$): $RSS = 1,098,908.5656$
- Model B ($p=8$, adds **s6**): $RSS = 1,089,717.0568$

(b) C_p , AIC, BIC and selections

Model	p	RSS	C_p	AIC ^{red}	BIC ^{red}
A	7	1,098,908.5656	8.1163	2974.6403	3005.9483
B	8	1,089,717.0568	7.0874	2973.5325	3008.7540

Selected: $C_p \rightarrow$ **Model B**, AIC $\rightarrow$ **Model B**, BIC $\rightarrow$ **Model A**.

(c) Why they can disagree; selection $\neq$ truth

Adding **s6** drops RSS by $1,098,908.5656 - 1,089,717.0568 = 9,191.5088$, at the cost of one extra parameter. C_p and AIC both charge a penalty equivalent to **2** extra “units” per additional parameter (2 in C_p ’s RSS-scaled units after dividing by $\hat{\sigma}^2$, and 2 directly in AIC^{red}’s log-RSS units), and here the RSS drop from **s6** is large enough to clear that bar - both go down when **s6** is added (C_p : $8.1163 \rightarrow 7.0874$; AIC: $2974.6403 \rightarrow 2973.5325$), so both prefer Model B.

BIC instead charges $\log(n) = \log(370) \approx 5.914$ per extra parameter - roughly **3 $\times$ steeper** than AIC’s fixed penalty of 2. The same **s6**-driven improvement that was “worth it” under a penalty of 2 is not worth it under a penalty of 5.914, so BIC^{red} actually goes *up* when **s6** is added ($3005.9483 \rightarrow 3008.7540$), and BIC prefers the smaller Model A. This is exactly the asymptotic distinction between the two criteria: AIC’s penalty stays fixed at 2 regardless of n , while BIC’s penalty grows with $\log(n)$, so BIC increasingly favors parsimony as the sample size grows, and the two criteria will diverge whenever an added predictor gives a “medium-sized” improvement - good enough to survive AIC’s lighter penalty but not BIC’s heavier one.

Selecting a model only tells you it fit *this particular penalized criterion* best among the *candidates you tried* (here, just A vs. B) - it says nothing about whether the linear functional form, the specific ten predictors, or the constant-variance/normality assumptions match the actual mechanism generating **y**. A different criterion (as shown here), a different candidate set, or a different sample could easily select a different “best” model from the same underlying data-generating process.

Question 5: Validation and final test data

(a) Where the analyst’s procedure goes wrong

Naive procedure results: best at $m = 9$, reported “test MSE” = 2369.3445.

Write-up: The misuse happens at **step 2** - evaluating *every* candidate model ($m = 0, \dots, 10$) on the final test set. The moment those eleven test-MSE numbers are computed and compared, the final test set has been used to *choose* the model (step 3 acts on it), even though it was never used to *fit* coefficients. The reported 2369.3445 is the minimum of eleven noisy MSE estimates - you’re implicitly asking “which of these eleven draws happened to look best against this particular 72-row test set,” which is a biased-low estimate of that model’s true future error, for the same reason a single test-MSE curve in Question 1 jumped around: it’s one noisy realization, and taking a *minimum* over several such realizations systematically favors whichever one got a lucky draw. The more candidates compared, the worse this optimism gets.

(b) Five-fold CV version

CV-selected $m = 9$ (CV MSE = 3126.3628, the lowest among the 11 candidates). Refitting that model on the full training data and evaluating once on the untouched final test set gives 2369.3445.

Write-up: Predictor count m is chosen by minimizing the CV MSE averaged across the 5 folds - each fold trains on the other 4/5 of the *training* data and validates on the held-out 1/5, repeated across all 11 values of m so every candidate is compared on the same folds. Once $m = 9$ is selected, that model is **refit on the entire training set** (rows 1–370) - cross-validation is only used to pick m , never to produce the final coefficients. The final test set (rows 371–442) is touched exactly once, at the very end, to report the prediction error of that single, already-fixed model.

The fact that $m = 9$ won under both procedures - and that the two “test MSE” numbers happen to be numerically identical (2369.3445) - is not a coincidence that vindicates the naive approach; it’s just this particular dataset/split landing on the same answer either way. Notice the CV MSE itself (3126.3628) is considerably *higher* than any of the naive procedure’s eleven test-set numbers, including the “worse” candidates - that gap is the optimism from part (a) made visible: the naive test-MSE values are all deflated because the test set was used for selection, while the CV MSE is an honest, selection-free estimate. Had the split been slightly different, the naive and CV procedures could easily have selected different m ’s, and the naive approach would have overstated its winner’s performance.

(c) What’s salvageable once the test set has been peeked at

Once all eleven test-MSE values have been examined, you can still honestly report the *relative ranking* - that $m = 9$ did best on that particular test set, and by roughly how much over the runner-up - as long as you’re explicit that 2369.3445 is no longer a valid, unbiased estimate of $m = 9$ ’s future prediction error (it’s now an optimistic, selection-tainted number, per part (a)); the CV MSE of 3126.3628 is the more trustworthy estimate of what to expect going forward). To get a genuinely fresh, honest final performance number for the selected model, you’d need a new batch of previously unseen data - or a held-out portion set aside *before* any peeking - on which to evaluate the one already-chosen $m = 9$ model exactly once, with no further comparison or selection happening against it.